

TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN, ĐHQG - TPHCM
KHOA CÔNG NGHỆ THÔNG TIN



Bài tập: Linux

Thực hành Linux

Học Phần: Nhập Môn DevOps

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1 Bài làm

1.1. User & Group

```
akira@Huy:~$ sudo useradd -m -s /bin/bash peter
akira@Huy:~$ sudo useradd -m -s /bin/bash alice
akira@Huy:~$ sudo passwd peter
New password:
Retype new password:
passwd: password updated successfully
akira@Huy:~$ sudo passwd alice
New password:
Retype new password:
passwd: password updated successfully
akira@Huy:~$ sudo useradd -m -d /home/bob_dir -s /bin/sh bob
akira@Huy:~$ sudo passwd bob
New password:
Retype new password:
Sorry, passwords do not match.
passwd: Authentication token manipulation error
passwd: password unchanged
akira@Huy:~$ sudo passwd bob
New password:
Retype new password:
passwd: password updated successfully
akira@Huy:~$ sudo usermod -l bigman bob -m -d /home/bigman_dir
akira@Huy:~$ sudo userdel -r alice
userdel: alice mail spool (/var/mail/alice) not found
akira@Huy:~$ |
```

Hình 1: Những câu lệnh của bài 1 thuộc phần 1

Các user là bob, alice, bob đều có password là trùng với tên của user để phục vụ cho bài tập dễ hơn.

```
akira@Huy:~$ sudo groupadd tech
akira@Huy:~$ sudo groupadd marketing
akira@Huy:~$ sudo usermod -aG tech peter
akira@Huy:~$ sudo usermod -aG tech bob
usermod: user 'bob' does not exist
akira@Huy:~$ sudo usermod -aG tech bigman
akira@Huy:~$ sudo usermod -aG marketing bigman
akira@Huy:~$ groups bigman
bigman : bob tech marketing
akira@Huy:~$ groups peter
peter : peter tech
akira@Huy:~$ sudo groupdel marketing
akira@Huy:~$ groups bigman
bigman : bob tech
```

Hình 2: Hình ảnh các câu lệnh được thực thi ở bài 2

1.2. Phân quyền file và thư mục

```
akira@Huy:~$ sudo su peter
[sudo] password for akira:
peter@Huy:/home/akira$ cd ..
peter@Huy:/home$ ls
akira  bigman_dir  peter
peter@Huy:/home$ cd peter
peter@Huy:~$ touch pratice.txt
peter@Huy:~$ chmod u=rw, g=r, o=r pratice.txt
chmod: invalid mode: 'u=rw,'
Try 'chmod --help' for more information.
peter@Huy:~$ chmod u=rw,g=r,o=r pratice.txt
peter@Huy:~$ ls -al | grep pratice.txt
-rw-r--r-- 1 peter peter    0 Jan 31 20:43 pratice.txt
peter@Huy:~$ chmod u=r pratice.txt
peter@Huy:~$ ls -al | grep pratice.txt
-r--r--r-- 1 peter peter    0 Jan 31 20:43 pratice.txt
peter@Huy:~$ |
```

Hình 3: Hình ảnh các câu lệnh đã thực thi cho bài tập 1

```

akira@Huy:~$ sudo su peter
[sudo] password for akira:
peter@Huy:/home/akira$ cd ..
peter@Huy:/home$ cd peter
peter@Huy:~$ cat <<EOF>>hello.sh
> #!/bin/bash
> echo 'hello'
> EOF
peter@Huy:~$ chmod 744 hello.sh
peter@Huy:~$ ls -al | grep hello.sh
-rwxr--r-- 1 peter peter  24 Jan 31 21:05 hello.sh
peter@Huy:~$ chmod u=rw hello.sh
peter@Huy:~$ ls -al | grep hello.sh
-rw-r--r-- 1 peter peter  24 Jan 31 21:05 hello.sh
peter@Huy:~$ ./hello.sh
bash: ./hello.sh: Permission denied
peter@Huy:~$ sudo ./hello.sh
[sudo] password for peter:
peter is not in the sudoers file.
peter@Huy:~$ ls
hello.sh  pratice.txt
peter@Huy:~$ chmod g=r, o=r
chmod: invalid mode: 'g=r,'
Try 'chmod --help' for more information.
peter@Huy:~$ chmod g=r,o=r
chmod: missing operand after 'g=r,o=r'
Try 'chmod --help' for more information.
peter@Huy:~$ chmod g=r,o=r hello.sh
peter@Huy:~$ ls -al | grep hello.sh
-rw-r--r-- 1 peter peter  24 Jan 31 21:05 hello.sh
peter@Huy:~$ |

```

Hình 4: Những câu lệnh thực thi trong bài 2

1.3. Quản lý xuất nhập file

```

akira@Huy:~$ touch myfile.txt
akira@Huy:~$ echo "Hello, Linux!" > myfile.txt
akira@Huy:~$ cat myfile.txt
Hello, Linux!
akira@Huy:~$ echo "Welcome to Linux file management." >> myfile.txt
akira@Huy:~$ cat myfile.txt
Hello, Linux!
Welcome to Linux file management.
akira@Huy:~$ wc -l myfile.txt
2 myfile.txt
akira@Huy:~$ wc -w myfile.txt
7 myfile.txt
akira@Huy:~$ cp myfile.txt myfile_backup.txt
akira@Huy:~$ ls
Project-HdH SystemCall Xv6-HCMUS-HK2-24-25 myfile.txt myfile_backup.txt xv6-labs-2023 xv6-labs-2024
akira@Huy:~$ mkdir tmp
akira@Huy:~$ mv myfile_backup.txt tmp/
akira@Huy:~$ cd tmp
akira@Huy:~/tmp$ ls
myfile_backup.txt
akira@Huy:~/tmp$ cd .
akira@Huy:~/tmp$ cd ..
akira@Huy:~$ rm myfile.txt
akira@Huy:~$ ls
Project-HdH SystemCall Xv6-HCMUS-HK2-24-25 tmp xv6-labs-2023 xv6-labs-2024

```

Hình 5: Những câu lệnh được thực hiện từ bài 1 đến bài 9

```

akira@Huy:~$ ls -al
total 96
drwxr-x--- 14 akira akira 4096 Jan 31 21:27 .
drwxr-xr-x  5 root  root  4096 Jan 31 20:32 ..
lrwxrwxrwx  1 akira akira   24 Oct 17 19:31 .aws -> /mnt/c/Users/ASUS99/.aws
lrwxrwxrwx  1 akira akira   26 Oct 17 19:31 .azure -> /mnt/c/Users/ASUS99/.azure
-rw-----  1 akira akira 12536 Jan 31 19:21 .bash_history
-rw-r-----  1 akira akira  220 Feb 14 2025 .bash_logout
-rw-r-----  1 akira akira 3771 Feb 14 2025 .bashrc
drwx-----  4 akira akira 4096 Feb 14 2025 .cache
drwxr-xr-x  5 akira akira 4096 Oct 17 19:31 .docker
drwxr-xr-x  3 akira akira 4096 Feb 14 2025 .dotnet
-rw-r-----  1 akira akira  62 Feb 25 2025 .gitconfig
drwxr-xr-x  2 akira akira 4096 Feb 24 2025 .landscape
-rw-----  1 akira akira  20 Mar  3 2025 .lessht
drwxr-xr-x  3 akira akira 4096 Mar  2 2025 .local
-rw-r-----  1 akira akira   0 Jan 31 14:44 .motd_shown
-rw-r-----  1 akira akira  807 Feb 14 2025 .profile
-rw-r-----  1 akira akira   0 Feb 14 2025 .sudo_as_admin_successful
drwxr-xr-x  5 akira akira 4096 Feb 14 2025 .vscode-server
-rw-r-----  1 akira akira  183 Jan 31 19:16 .wget-hsts
drwxr-xr-x  8 akira akira 4096 Apr  8 2025 Project-HdH
drwxr-xr-x  7 akira akira 4096 Apr  8 2025 SystemCall
drwxr-xr-x  8 akira akira 4096 Mar  3 2025 Xv6-HCMUS-HK2-24-25
drwxrwxr-x  2 akira akira 4096 Jan 31 21:27 tmp
drwxr-xr-x  9 akira akira 4096 Apr  8 2025 xv6-labs-2023
drwxr-xr-x  8 akira akira 4096 Mar 11 2025 xv6-labs-2024

```

Hình 6: Bài 9

```
akira@Huy:~$ grep "Linux" /tmp/myfile_backup.txt
grep: /tmp/myfile_backup.txt: No such file or directory
akira@Huy:~$ cd tmp
akira@Huy:~/tmp$ grep "Linux" /tmp/myfile_backup.txt
grep: /tmp/myfile_backup.txt: No such file or directory
akira@Huy:~/tmp$ grep "Linux" myfile_backup.txt
Hello, Linux!
Welcome to Linux file management.
akira@Huy:~/tmp$ cd ..
akira@Huy:~$ grep "Linux" ./tmp/myfile_backup.txt
Hello, Linux!
Welcome to Linux file management.
akira@Huy:~$ mv ./tmp/myfile_backup.txt ./tmp/linuxfile.txt
akira@Huy:~$ cd tmp
akira@Huy:~/tmp$ ls
linuxfile.txt
akira@Huy:~/tmp$ cd ..
akira@Huy:~$ mkdir backup
akira@Huy:~$ mv ./tmp/linuxfile.txt backup/
akira@Huy:~$ ls
Project-HdH  SystemCall  Xv6-HCMUS-HK2-24-25  backup  tmp  xv6-labs-2023  xv6-labs-2024
akira@Huy:~$ cd backup
akira@Huy:~/backup$ ls
linuxfile.txt
akira@Huy:~/backup$ cd ..
akira@Huy:~$ less ./backup/linuxfile.txt
akira@Huy:~$ less ./backup/linuxfile.txt
akira@Huy:~$ less ./backup/linuxfile.txt
```

Hình 7: Những câu lệnh được sử dụng từ bài 10 đến bài 13

1.4. Quản lý gói phần mềm

```
akira@Huy:~$ sudo apt update
[sudo] password for akira:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1410 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1717 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [229 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [9820 B]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [925 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [209 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.7 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [317 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2369 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.0 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1525 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [312 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [543 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:23 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [536 B]
Get:24 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [28.8 kB]
Get:25 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6492 B]
Get:26 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:27 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Get:28 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.8 kB]
Get:29 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2536 kB]
Get:30 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [580 kB]
Get:31 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:32 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [556 B]
Get:33 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [32.1 kB]
Get:34 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [6816 B]
Get:35 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:36 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [496 B]
Get:37 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.4 kB]
Get:38 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7300 B]
Get:39 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
```

Hình 8: Lệnh câu 1

Sau khi thực hiện câu lệnh “sudo apt update”, hệ thống sẽ duyệt in ra thông báo là “158 packages can be upgraded”. Run “apt list –upgradable to see them”.


```
akira@Huy:~$ sudo apt install vsftpd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-intel1 libpciaccess0 libsensors-config libsensors5
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libwrap0 ssl-cert
The following NEW packages will be installed:
  libwrap0 ssl-cert vsftpd
0 upgraded, 3 newly installed, 0 to remove and 158 not upgraded.
Need to get 185 kB of archives.
After this operation, 489 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libwrap0 amd64 7.6.q-33 [47.9 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 vsftpd amd64 3.0.5-0ubuntu3.1 [120 kB]
Fetched 185 kB in 2s (75.6 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libwrap0:amd64.
(Reading database ... 51472 files and directories currently installed.)
Preparing to unpack .../libwrap0_7.6.q-33_amd64.deb ...
Unpacking libwrap0:amd64 (7.6.q-33) ...
Selecting previously unselected package ssl-cert.
Preparing to unpack .../ssl-cert_1.1.2ubuntu1_all.deb ...
Unpacking ssl-cert (1.1.2ubuntu1) ...
Selecting previously unselected package vsftpd.
Preparing to unpack .../vsftpd_3.0.5-0ubuntu3.1_amd64.deb ...
Unpacking vsftpd (3.0.5-0ubuntu3.1) ...
Setting up libwrap0:amd64 (7.6.q-33) ...
Setting up ssl-cert (1.1.2ubuntu1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ssl-cert.service → /usr/lib/systemd/system/ssl-cert.service.
Setting up vsftpd (3.0.5-0ubuntu3.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/vsftpd.service → /usr/lib/systemd/system/vsftpd.service.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
```

Hình 9: Câu lệnh thực hiện bài 2

```
akira@Huy:~$ apt search net-tools
Sorting... Done
Full Text Search... Done
ata-tools/noble 1:2.5.1-5.1build1 amd64
  Base programs for ATM in Linux, the net-tools for ATM

ddnet-tools/noble 16.4-1.2build5 amd64
  Tools for DQNet

hobbit-plugins/noble 20230301 all
  plugins for the Xymon network monitor

iproute2/noble-updates 6.1.0-1ubuntu6.2 amd64 [upgradable from: 6.1.0-1ubuntu6]
  networking and traffic control tools

net-tools/noble-updates,noble-security 2.10-0.1ubuntu4.4 amd64
  NET-3 networking toolkit

akira@Huy:~$ apt show vsftpd
Package: vsftpd
Version: 3.0.5-0ubuntu3.1
Priority: extra
Section: net
Origin: Ubuntu
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Original-Maintainer: Keng-Yu Lin <kengyu@debian.org>
Bugs: https://bugs.launchpad.net/ubuntu/+filebug
Installed-Size: 312 kB
Provides: ftp-server
Depends: debconf (>= 0.5) | debconf-2.0, libc6 (>= 2.34), libcap2 (>= 1:2.10), libpam0g (>= 0.99.7.1), libssl3t64 (>= 3.0.0), libwrap0 (>= 7.6-4-), adduser, libpam-modules, lsb-base
(0-6), netbase, procps, sysvinit-utils (>= 2.96)
Recommends: logrotate, ssl-cert
Conflicts: ftp-server
Replaces: ftp-server
Homepage: http://vsftpd.beasts.org/
Download-Size: 120 kB
APT-Manual-Installed: yes
APT-Sources: http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages
Description: lightweight, efficient FTP server written for security
  This package provides the "Very Secure FTP Daemon", written from
  the ground up with security in mind.
  .
  It supports both anonymous and non-anonymous FTP access, PAM authentication,
  bandwidth limiting, and the Linux sendfile() facility.

N: There is 1 additional record. Please use the '-a' switch to see it
```

Hình 10: Câu lệnh thực hiện bài 3 và bài 4

```
akira@Huy:~$ sudo apt remove vsftpd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-intel1 libpciaccess0 libsensors-config libsensors5 libwrap0 ssl-cert
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  vsftpd
0 upgraded, 0 newly installed, 1 to remove and 158 not upgraded.
After this operation, 312 kB disk space will be freed.
Do you want to continue? [Y/n] Y
(Reading database ... 51549 files and directories currently installed.)
Removing vsftpd (3.0.5-0ubuntu3.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
```

Hình 11: Bài 5

```
akira@Huy:~$ dpkg -l | grep python
ii  libpython3-stdlib:amd64      3.12.3-0ubuntu2      amd64      interactive high-level object-oriented language (default python3 version)
ii  libpython3.12-minimal:amd64  3.12.3-1ubuntu0.8    amd64      Minimal subset of the Python language (version 3.12)
ii  libpython3.12-stdlib:amd64   3.12.3-1ubuntu0.8    amd64      Interactive high-level object-oriented language (standard library, version 3.12)
ii  libpython3.12t64:amd64       3.12.3-1ubuntu0.8    amd64      Shared Python runtime library (version 3.12)
ii  python3-apt-common           2.7.7ubuntu4         all        Python interface to libapt-pkg (locales)
ii  python3-babel-localedata     2.10.3-3build1       all        tools for internationalizing Python applications - locale data files
ii  python3                      3.12.3-0ubuntu2       amd64      interactive high-level object-oriented language (default python3 version)
ii  python3-apport               2.28.1-0ubuntu3.8    all        Python 3 library for Apport crash report handling
ii  python3-apt                  2.7.7ubuntu4         amd64      Python 3 interface to libapt-pkg
ii  python3-attr                 23.2.0-2             all        Attributes without boilerplate (Python 3)
ii  python3-automat              22.10.0-2            all        Self-service finite-state machines for the programmer on the go
ii  python3-babel               2.10.3-3build1       all        tools for internationalizing Python applications - Python 3.x
ii  python3-bcrypt              3.2.2-1build1        amd64      password hashing library for Python 3
ii  python3-blinker              1.7.0-1              all        Fast, simple object-to-object and broadcast signaling (Python3)
ii  python3-certifi              2023.11.17-1         all        root certificates for validating SSL certs and verifying TLS hosts (python3)
ii  python3-cffi-backend:amd64   1.16.0-2build1       amd64      Foreign Function Interface for Python 3 calling C code - runtime
ii  python3-chardet              5.2.0+dfsg-1         all        Universal Character Encoding Detector (Python3)
ii  python3-click                8.1.6-2              all        Wrapper around optparse for command line utilities - Python 3.x
ii  python3-colorama             0.4.6-4              all        Cross-platform colored terminal text in Python - Python 3.x
ii  python3-commandnotfound      23.04.0              all        Python 3 bindings for command-not-found.
ii  python3-configobj            5.0.8-3              all        simple but powerful config file reader and writer for Python 3
ii  python3-constantly           23.10.4-1            all        Symbolic constants in Python
ii  python3-cryptography         41.0.7-4ubuntu0.1    amd64      Python library exposing cryptographic recipes and primitives (Python 3)
ii  python3-dbus                 1.3.2-5build3        amd64      simple interprocess messaging system (Python 3 interface)
ii  python3-debconf              1.5.86ubuntu1        all        interact with debconf from Python 3
ii  python3-distutils             1.9.0-1              all        Linux OS platform information API
ii  python3-distutils-info       1.7build1            all        information about distributions' releases (Python 3 module)
ii  python3-distupgrader         1:24.04.26           all        manage release upgrades
ii  python3-gdbm:amd64           3.12.3-0ubuntu1      amd64      GNU dbm database support for Python 3.x
ii  python3-gi                   3.48.2-1             amd64      Python 3 bindings for GObject-introspection libraries
ii  python3-hancrest              2.1.0-1              all        Hancrest framework for matcher objects (Python 3)
ii  python3-httplib2              0.20.4-3             all        comprehensive HTTP client library written for Python3
ii  python3-hyperlink            21.0.0-5             all        Immutable, Pythonic, correct URLs.
ii  python3-idna                  3.6-2ubuntu0.1       all        Python IDNA2008 (RFC 5891) handling (Python 3)
```

Hình 12: Bài 6

```
akira@Huy:~$ apt depends vim
vim
Depends: vim-common (= 2:9.1.0016-1ubuntu7.9)
Depends: vim-runtime (= 2:9.1.0016-1ubuntu7.9)
Depends: libacl1 (>= 2.2.23)
Depends: libc6 (>= 2.38)
Depends: libgpm2 (>= 1.20.7)
Depends: libpython3.12t64 (>= 3.12.1)
Depends: libselinux1 (>= 3.1~)
Depends: libsodium23 (>= 1.0.14)
Depends: libtinfo6 (>= 6)
Suggests: <ctags>
Suggests: exuberant-ctags
Suggests: universal-ctags
Suggests: vim-doc
Suggests: vim-scripts
akira@Huy:~$ apt --fix-broken install
E: Could not open lock file /var/lib/dpkg/lock-frontent - open (13: Permission denied)
E: Unable to acquire the dpkg frontend lock (/var/lib/dpkg/lock-frontent), are you root?
akira@Huy:~$ sudo apt --fix-broken install
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-intel1 libpciaccess0 libsensors-config libsensors5 libwrap0 ssl-cert
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 158 not upgraded.
```

Hình 13: Bài 7 và bài 8

```
akira@Huy:~$ sudo apt autoremove
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be REMOVED:
  libdrm-intel1 libpciaccess0 libsensors-config libsensors5 libwrap0 ssl-cert
0 upgraded, 0 newly installed, 6 to remove and 156 not upgraded.
After this operation, 522 kB disk space will be freed.
Do you want to continue? [Y/n] Y
(Reading database ... 51498 files and directories currently installed.)
Removing libdrm-intel1:amd64 (2.4.122-1~ubuntu0.24.04.1) ...
Removing libpciaccess0:amd64 (0.17-3build1) ...
Removing libsensors5:amd64 (1:3.6.0-9build1) ...
Removing libsensors-config (1:3.6.0-9build1) ...
Removing libwrap0:amd64 (7.6.q-33) ...
Removing ssl-cert (1.1.2ubuntu1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
```

Hình 14: Bài 10

1.5. Quản lý gói dịch vụ

```
akira@Huy:~$ sudo apt install nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 156 not upgraded.
Need to get 564 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.5 [43.4 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.5 [520 kB]
Fetched 564 kB in 2s (234 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 51458 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.5_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.5) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.5_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.5) ...
Setting up nginx-common (1.24.0-2ubuntu7.5) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.5) ...
* Upgrading binary nginx
Processing triggers for man-db (2.12.0-4build2) ...
```

Hình 15: Bài 1a

```
akira@Huy:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-01-31 21:52:51 +07; 16s ago
     Docs: man:nginx(8)
  Process: 2709 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
  Process: 2710 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 2745 (nginx)
      Tasks: 9 (limit: 4591)
    Memory: 6.4M (○)
   CGroup: /system.slice/nginx.service
           └─2745 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─2747 "nginx: worker process"
               └─2748 "nginx: worker process"
                 └─2749 "nginx: worker process"
                   └─2750 "nginx: worker process"
                     └─2751 "nginx: worker process"
                       └─2752 "nginx: worker process"
                         └─2753 "nginx: worker process"
                           └─2754 "nginx: worker process"

Jan 31 21:52:51 Huy systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jan 31 21:52:51 Huy systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
```

Hình 16: Bài 1b

```
akira@Huy:~$ systemctl start nginx
===== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units =====
Authentication is required to start 'nginx.service'.
Authenticating as:,,, (akira)
Password:
===== AUTHENTICATION COMPLETE =====
akira@Huy:~$ systemctl stop nginx
===== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units =====
Authentication is required to stop 'nginx.service'.
Authenticating as:,,, (akira)
Password:
===== AUTHENTICATION COMPLETE =====
akira@Huy:~$ |
```

Hình 17: Bài 1c

```
akira@Huy:~$ systemctl is-enabled nginx
enabled
akira@Huy:~$ sudo systemctl disable nginx
[sudo] password for akira:
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable nginx
Removed "/etc/systemd/system/multi-user.target.wants/nginx.service".
akira@Huy:~$ systemctl is-enabled nginx
disabled
akira@Huy:~$ systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
Reload daemon failed: Interactive authentication required.
update-rc.d: error: Permission denied
akira@Huy:~$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service
akira@Huy:~$ systemctl is-enabled nginx
enabled
```

Hình 18: Bài 2

```
akira@Huy:~$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
akira@Huy:~$ cat /var/log/nginx/access.log
::1 - - [02/Feb/2026:15:32:46 +0700] "GET / HTTP/1.1" 200 615 "-" "curl/8.5.0"
akira@Huy:~$ cat /var/log/nginx/error.log
2026/01/31 21:52:51 [notice] 2745#2745: using inherited sockets from "5;6;"
akira@Huy:~$ tail -f /var/log/nginx/error.log
tail: invalid option -- '/'
Try 'tail --help' for more information.
akira@Huy:~$ tail -f /var/log/nginx/error.log
2026/01/31 21:52:51 [notice] 2745#2745: using inherited sockets from "5;6;"
^C
```

Hình 19: Bài 3,4,5

1.6. Quản lý tác vụ tự động bằng Batch Script

```
GNU nano 7.2
#!/bin/bash
echo "Hello World"
```

Hình 20: Bài 1

```
#!/bin/bash
echo "Current time: $(date)"
```

Hình 21: Bài 2

```
#!/bin/bash
echo "Vui long nhap so dau tien"
read num1
echo "Vui long nhap so thu hai"
read num2
sum=$((num1+num2))
echo "Tong hai so la: $sum"
```

Hình 22: Bài 3

```
#!/bin/bash
echo "Nhap file can kiem tra: "
read file
if [ -f "$file" ]; then
    echo "tap tin $file ton tai"
else
    echo "tap tin $file khong ton tai"
fi
```

Hình 23: Bài 4

```
GNU nano 7.2
#!/bin/bash
ls -l|
```

Hình 24: Bài 5

```
GNU nano 7.2
#!/bin/bash
for i in {1..10}
do
    echo $i
done|
```

Hình 25: Bài 6

```
GNU nano 7.2
#!/bin/bash
for i in {1..100}
do
    echo $i
done|
```

Hình 26: Bài 7


```
#!/bin/bash
echo "Nhap file can dem so dong: "
read file
if [ -f "$file" ]; then
    line_count=$(wc -l < "$file")
    echo "So dong trong $file la: $line_count"
else
    echo "Tap tin $file khong ton tai."
fi
```

Hình 27: Bài 8

```
GNU nano 7.2
#!/bin/bash
echo "Nhap ten file can sao luu: "
read filename
if [ -f "$filename" ]; then
    backup_name="$filename.$(date +%Y%m%d%M%S).bak"
    cp $filename $backup_name
    echo "Da tao ban sao luu: $backup_name"
else
    echo "Tap tin $file khong ton tai."
fi
```

Hình 28: Bài 9

```
GNU nano 7.2
#!/bin/bash
echo "Script dang chay vao luc: $(date)" >> ~/output.log
```

Hình 29: Bài 10a

```
akira@Huy:~$ chmod +x ~/run_script.sh
akira@Huy:~$ crontab -e
crontab: installing new crontab
akira@Huy:~$ cat ~/output.log
Script đang chạy vào lúc: Mon Feb  2 16:48:01 +07 2026
akira@Huy:~$ |
```

Hình 30: Bài 10b

```
GNU nano 7.2
#!/bin/bash
backup_dir=~/.backup
source_dir=~/.myfolder
mkdir -p $backup_dir
tar -czf $backup_dir/backup_$(date +%Y%m%d).tar.gz -C $source_dir
|
```

Hình 31: Bài 11a

```
#
# m h dom mon dow   command
*/2 * * * * /bin/bash /home/akira/run_script.sh
0 2 * * * /bin/bash ~/backup.sh|
```

Hình 32: Bài 11b

2 Tài liệu tham khảo